



Genesis Biotech Inc.

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Anti-Coat Protein of Fish Nervous Necrosis virus, Rabbit-Polyclonal Antibody

Catalog No. GB-10062

Quantity: 250 μ l

Antigen species: Fish NNV

Reactivity: Fish NNV

Applications tested: ELISA

Host species: rabbit

Type: polyclonal antibody **Form:** lyophilized antiserum

Target description

Viral nervous necrosis (VNN) is a worldwide disease among marine fishes. Fish nervous necrosis virus (NNV) causes high mortality and considerable economic damage to the aquaculture industry. In Taiwan, VNN disease was first identified in 2 species of hatchery-reared grouper, *Epinephelus fuscogutatus* and *E. akaaya* in 1994. Since then, increasing mortalities have occurred among groupers *Epinephelus* spp., and also among European eels *Anguilla anguilla* L., yellow-wax pompano *Trachinotus falcatus*, firespot snapper *Lutjanus erythropterus* B., barramundi *Lates calcarifer*, cobias *Rachycentron canadum*, humpback groupers *Cromileptes altivelis* and Chinese catfish *Parasilurus asotus*.

Antigen

This polyclonal antibody was raised by immunizing rabbit with a synthetic peptide containing amino acids on the central domain of coat protein of NNV (aa 90-150) of grouper NNV. The antigen contained the epitope defined by Epitocscreen™ (Genesis Biotech Inc.).

Application

The antibody specificity was assayed by ELISA with the synthetic coat protein peptide antigen of fish NNV. The antibody titer is more than 10K for ELISA. The corresponding peptide antigen can also be bound with rabbit anti-recombinant GNNV coat protein polyclonal antibody. It has not been tested in the other applications. However, for the first testing, we recommend 1/5,000 dilution for ELISA, 1/1000 dilution for Western blot analysis (WB) of recombinant protein, 1/400 dilution for tissue extracts or cell lysates, 1/100 dilution for immunohistochemistry (IHC) staining on frozen cryosections, 1/50 dilution for IHC staining on paraffin embedded sections.

Ab dilution	prebleed	Antiserum
1:1K	0.063	1.631
1:10K	0.052	0.673
1:100K	0.046	0.093
Titer		~100 K

ELISA Procedure

Antigen is coated on EIA strips at 1 μ g per well. Add 200 μ l of blocking buffer and then wash wells with PBST buffer. Antiserum GB-10062 is diluted in series as 10^3 – 10^5 folds and added in separate wells. Incubate antibody for 1hr. Wash unbound antibodies and add anti-rabbit IgG-HRP conjugate. Wash the plates and add substrate to develop color for 5 min. Read absorbance (ABS) at 405 nm. Amount of color is directly proportional to the amount of antibodies. Antibody is positive at >2 folds of ABS of control/Pre-Immune serum.

Related Products

1. Anti-NNV coat protein rabbit pAb (GB-10063)
2. Anti-NNV coat protein rabbit pAb (GB-10064)
3. Anti-NNV particles rabbit pAb (PG-10002)

Storage

It is supplied as lyophilized serum of polyclonal antibody. Redissolve the lyophilized powder with 250 μ l sterile water will have the same concentration with serum. Store at 4°C for short term application. For long-term storage, aliquot and store at -20°C.

References

1. Chi SC, Shieh JR, Lin SJ. Genetic and antigenic analysis of beta nodaviruses isolated from aquatic organisms in Taiwan. Dis Aquat Organ. 2003 Aug 4;55(3):221-8.
2. Chi SC, Lin SC, Su HM, Hu WW. Temperature effect on nervous necrosis virus infection in grouper cell line and in grouper larvae. Virus Res. 1999 Sep;63(1-2):107-14.